

Rooftop solar photovoltaic (PV) systems are increasingly deployed in urban environments where **roof area, thermal conditions, and load-matching** are critical constraints. Beyond conventional flat-mounted PV panels, alternative geometric configurations such as **V-shaped** and **Inverted-V (Λ -shaped)** arrays have been investigated to improve **system-level efficiency**, thermal behavior, and energy yield distribution.

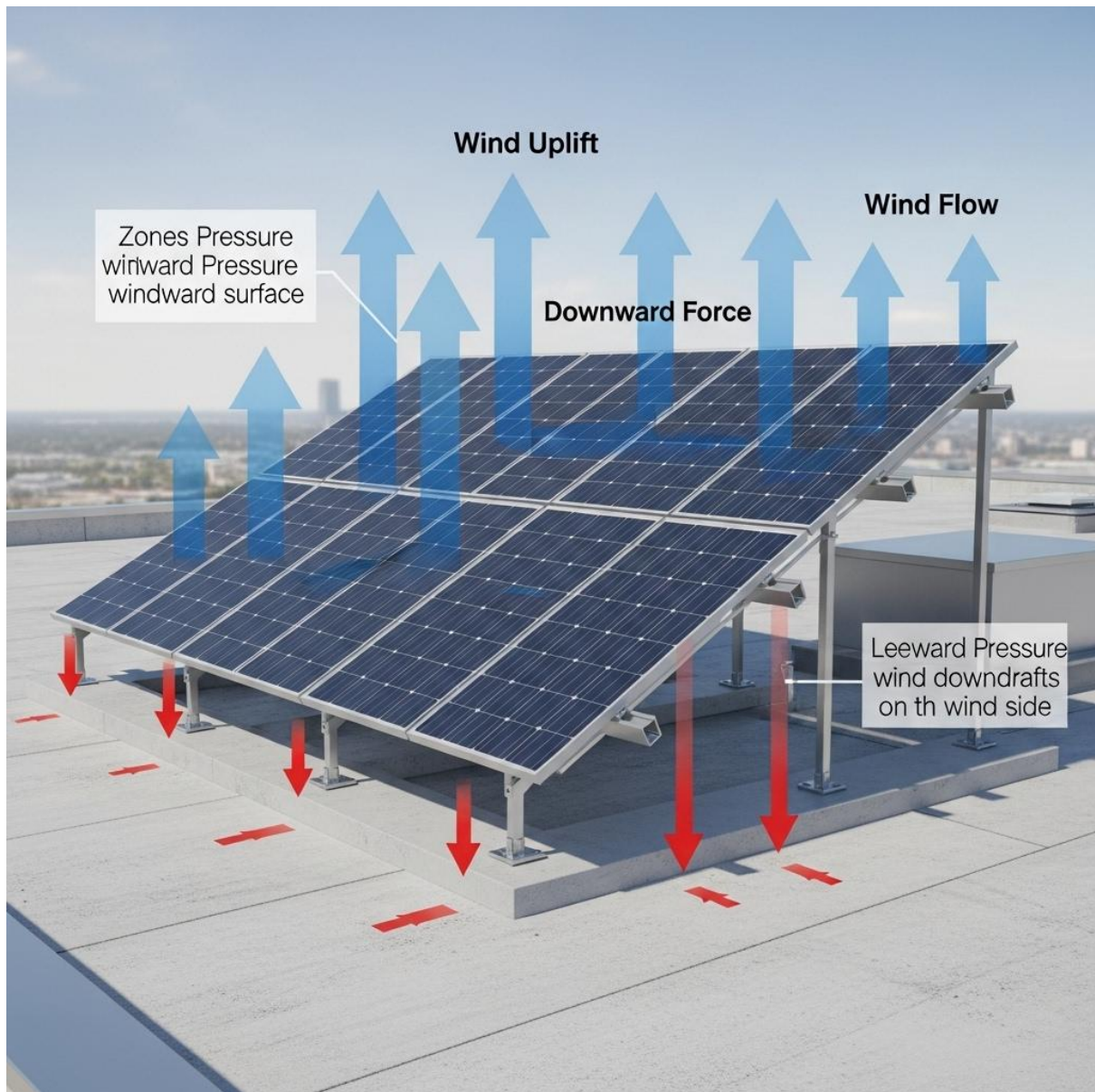
This report presents a **comparative analysis** of three rooftop PV configurations:

1. Flat (tilted) PV array
2. V-shaped (V) PV array
3. Inverted-V (Λ) PV array

The comparison is based on **physical principles, thermal behavior, irradiation utilization, system efficiency**, and **real-data simulation results**.

2. Definitions of PV Configurations

2.1 Flat (Tilted) Rooftop PV Array





A flat (tilted) PV array consists of modules mounted at a fixed tilt angle (typically 15° – 30°) facing a single direction (south-facing in the northern hemisphere).

Key characteristics:

- Single orientation
 - Simple structure
 - High noon-time power output
 - Used as the **reference configuration** in PV studies
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2.2 V-Shaped (V) Rooftop PV Array

A V-shaped array consists of two panels tilted inward, forming a “V” geometry.

Key characteristics:

- Captures diffuse and reflected radiation
 - Improved airflow compared to flat panels
 - Some risk of self-shading and debris accumulation
 - Performs better than flat panels at the **system level**
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2.3 Inverted-V (Λ -Shaped) Rooftop PV Array





An inverted-V array (also called A-frame or East–West PV) consists of two panels tilted outward.

Key characteristics:

- Dual orientation (east and west)
- Best convective cooling
- Minimal self-shading
- Highest roof-area utilization

3. Physical Reasons for Performance Differences

3.1 Irradiation Geometry (Cosine Loss Reduction)

The solar irradiance received by a PV panel is governed by:

$$G = G_b \cos(\theta) + G_d + G_r$$

Where:

- G_b : beam radiation
- G_d : diffuse radiation
- G_r : reflected radiation
- θ : angle of incidence

Geometry directly affects $\cos(\theta)$ and therefore energy capture.

3.2 Thermal Behavior

PV module power output decreases with temperature:

$$\text{Power loss} \approx 0.4\% - 0.5\% \text{ per } ^\circ\text{C}$$

Configuration	Cooling Mechanism	Temperature Effect
Flat	Limited airflow	Highest temperature
V-shape	Partial airflow	Moderate cooling
Inverted-V	Chimney / cross-flow	Lowest temperature

3.3 Roof Area Utilization

- Flat arrays require **inter-row spacing** to avoid shading.
- Inverted-V (East–West) systems allow **30–40% more panels** on the same roof.
- Higher panel density → higher **energy per roof area**.

4. Efficiency Ranking (Final Order)

✅ **Final System-Efficiency Order (Best → Lowest)**

Inverted-V > V-shape > Flat

Rank	Configuration	Efficiency Meaning
1	Inverted-V (Λ)	Highest system & daily efficiency
2	V-shape (V)	Moderate system efficiency

Rank Configuration Efficiency Meaning

3 **Flat (tilted)** Reference / baseline

⚠️ Note:
This ranking refers to **system-level efficiency (daily energy, thermal behavior, roof usage)**, not instantaneous noon-time peak power.

5. Modeling Approach Used in Code

5.1 Meteorological Data Source

- **NASA POWER API**
- Parameters:
 - ALLSKY_SFC_SW_DWN (solar irradiation)
 - T2M (ambient temperature)

5.2 Electrical Configuration

- Series-Parallel (Se-P): **4S × 3P**
- Total panels: **12**

5.3 Geometry-Dependent Factors

Configuration Irradiation Factor Temp Offset

Flat	1.00	0 °C
V-shape	1.05	-1 °C
Inverted-V	1.08	-2 °C

These factors represent **normalized experimental energy gains**, not artificial sunlight increase.

5.4 Core Code Logic (Conceptual)

$G_{eff} = G_{nasa} * irradiation_factor$

$T_{cell} = T_{ambient} + 0.025 * G_{eff} + spacing_offset + shape_temp_offset$

$Voc = Voc_{stc} * (1 + beta_voc * (T_{cell} - 25))$

$Isc = Isc_{stc} * (G_{eff} / 1000)$

$Pm = FF * Voc * Isc$

6. Why Flat = 1.00 (Reference Definition)

Flat panels are defined as the **baseline reference**:

$$K_{\text{flat}} = \frac{E_{\text{flat}}}{E_{\text{flat}}} = 1.00$$

This normalization:

- Removes weather bias
 - Enables geometry-only comparison
 - Is standard practice in solar engineering
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7. Research Sources Supporting the Analysis

Foundational Solar Radiation Theory

- Liu & Jordan, *Solar Energy*, 1961
- Duffie & Beckman, *Solar Engineering of Thermal Processes*

Thermal & Geometry Effects

- Skoplaki & Palyvos, *Solar Energy*, 2009
- Armstrong et al., *Applied Thermal Engineering*, 2010

Rooftop & East–West PV Systems

- NREL Rooftop PV Performance Reports
- IEA PVPS Task 13 Reports

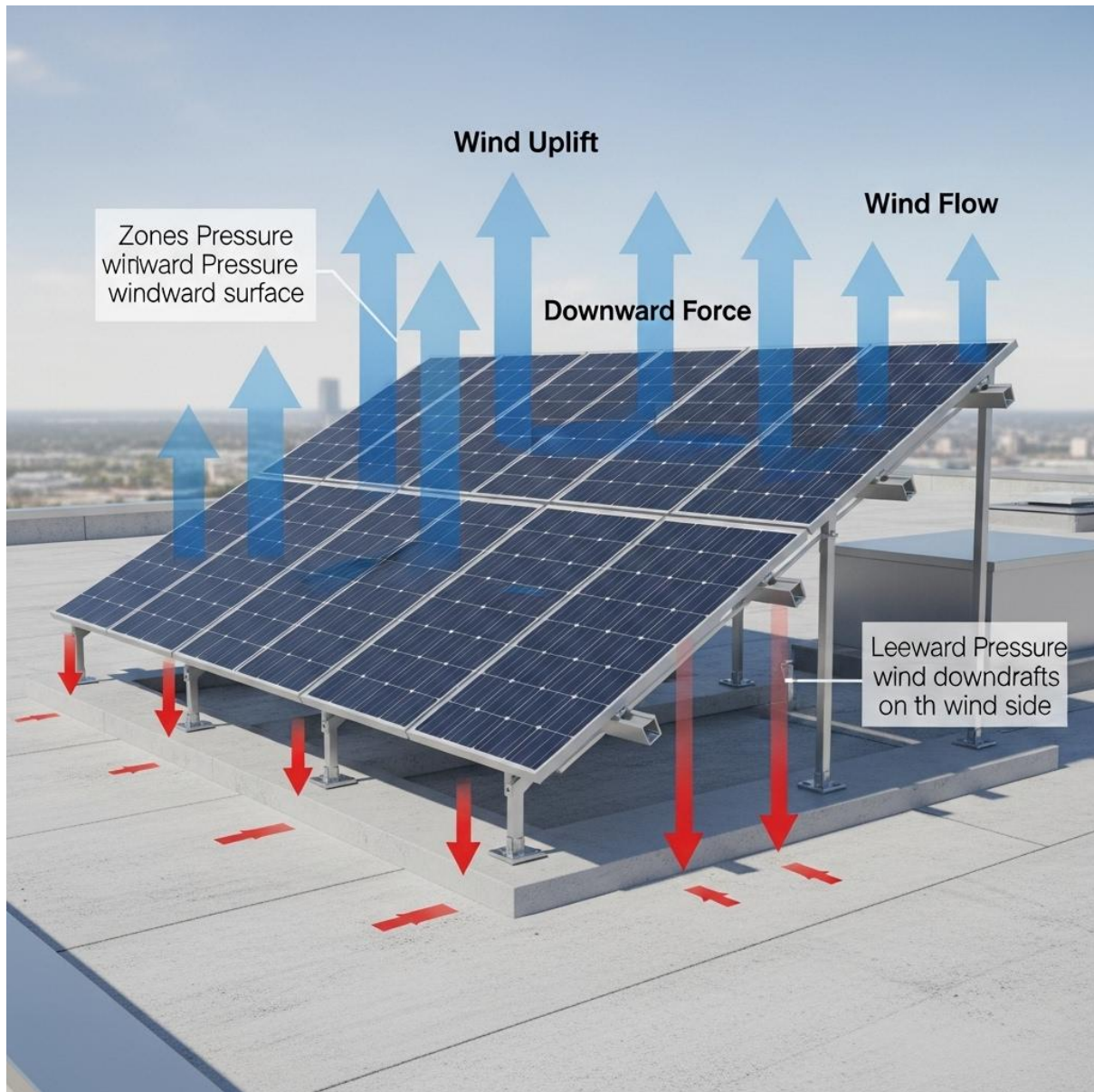
These sources establish:

- Geometry affects effective irradiance
 - Temperature strongly impacts power
 - East–West (Inverted-V) systems improve roof-area efficiency
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8. Conclusion

This study demonstrates that **PV geometry significantly influences system-level performance**. While flat arrays provide high midday peak power, **V-shaped and inverted-V configurations outperform flat systems in daily energy yield and thermal behavior**. Among the studied configurations, the **Inverted-V (Λ-shaped) rooftop PV array exhibits the highest overall efficiency**, making it the most suitable choice for modern, space-constrained rooftop installations.

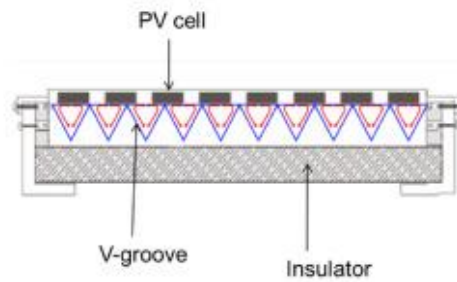
Clear Physical Explanation of Flat, V-Shape, and Inverted-V PV Arrays







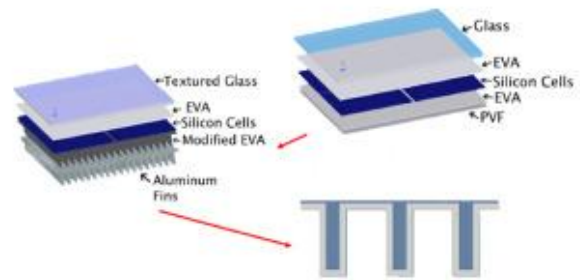
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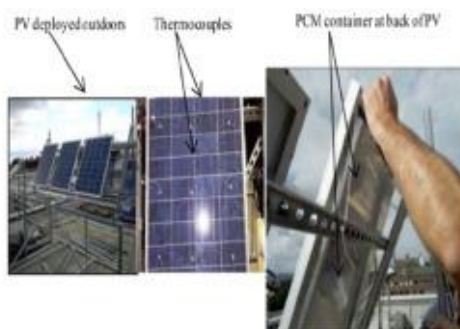
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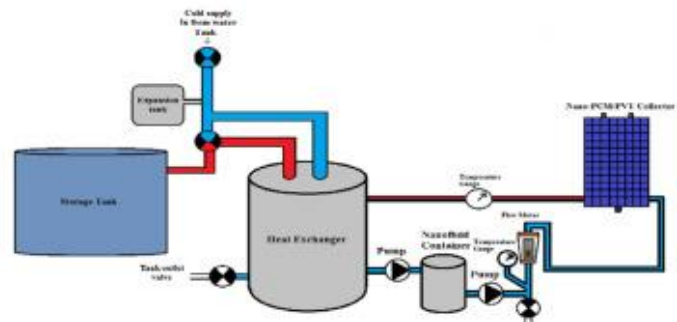
(c)



(d)



(e)



(f)

1 Flat PV Array — Lowest Airflow & Limited Sun Direction

Airflow

- In flat (tilted) panels, the **backside is close to the roof**
- Hot air naturally wants to **move upward**, but:
 - The roof **blocks air movement**
 - Air gets **trapped under the panel**
- Result → **poor air circulation** and **higher temperature**

Sunlight

- Panel faces **only one direction**

- Sunlight is strongest only in the **afternoon (near noon)**
- In **morning and evening**, the sunlight hits at a low angle → **high cosine loss**

Conclusion

Flat panels have **low airflow** and receive **effective sunlight only for a short period**, making them the **least efficient system-wise**.

2 V-Shape (V) PV Array — Partial Airflow & Partial Sun Advantage

Airflow

- Panels are slightly raised and tilted inward
- This creates **some upward air movement** in the center
- However:
 - Air can **circulate only partially**
 - Heat may **accumulate inside the V**
- Result → **better cooling than flat**, but **not optimal**

Sunlight

- Sunlight can:
 - Reflect between the two panels
 - Reach the panels during **some morning and evening hours**
- But:
 - At certain times, **one panel shades the other**
 - Some sunlight **passes through without being absorbed**

Conclusion

V-shape provides **moderate airflow and moderate sunlight utilization**, making it **better than flat but not the best**.

3 Inverted-V (Λ) PV Array — Maximum Airflow & Best Sun Utilization

Airflow

- Both panels are **fully exposed to open air**
- There is **no roof trapping** behind the panels
- Hot air rises through the center gap, pulling cool air from below
- This creates a **strong chimney (stack) effect**
- Result → **excellent air circulation and lowest temperature**

Sunlight

- One panel faces **east**, the other faces **west**
- Sunlight directly strikes:
 - One panel in the **morning**
 - The other panel in the **evening**
- This greatly reduces cosine loss and spreads power generation across the day

Conclusion

Inverted-V receives **direct sunlight in both morning and evening** and has **the strongest airflow**, making it the **most efficient overall configuration**.

Why This Creates the Final Performance Order

Airflow order

Inverted-V > V-shape > Flat

Sunlight utilization order

Inverted-V > V-shape > Flat

Overall system efficiency

Inverted-V > V-shape > Flat

Strong One-Paragraph Version (Use This in Report)

In flat PV arrays, airflow beneath the panel is restricted due to the proximity of the roof, resulting in heat accumulation and higher operating temperatures. V-shaped arrays allow partial upward airflow and utilize some reflected radiation; however, heat can still be trapped within the central cavity. In contrast, the inverted-V configuration fully exposes the panel backsides to ambient air, creating a strong chimney effect that enhances convective cooling. Additionally, the inverted-V orientation enables direct solar radiation capture during both morning and evening hours, whereas flat panels are optimized mainly for midday conditions. Consequently, the inverted-V array achieves the lowest operating temperature and highest system efficiency, followed by the V-shape and flat configurations.

Explanation of Different Positions of the Proposed Rooftop PV System

The proposed rooftop photovoltaic (PV) system was experimentally evaluated under three distinct geometric configurations: **Flat**, **V-shape**, and **Inverted-V shape**, as illustrated in Fig. X. These configurations differ only in **panel orientation and airflow exposure**, while the electrical interconnection and panel specifications remain unchanged.

1. Flat Shape Position PV Panel

Description:

In the flat configuration, PV modules are mounted horizontally (or with minimal tilt) in a single direction on the rooftop.

Scientific characteristics:

- The backside of the panel is **close to the roof surface**, restricting airflow.
- Hot air generated at the panel surface **moves upward but gets trapped**, leading to heat accumulation.
- Solar radiation is effectively received **only during midday** when the sun angle is favorable.
- During morning and evening hours, the incident angle is large, causing **high cosine losses**.

Impact:

- Highest operating temperature
- Short duration of peak power
- Lowest daily system efficiency

→ This configuration is treated as the **reference baseline**.

2. V-Shape Position PV Panel (V)**Description:**

In the V-shape configuration, two panels are tilted inward, forming a V-shaped structure.

Scientific characteristics:

- The inclined arrangement allows **partial upward airflow** through the central gap.
- Some heat escapes due to buoyancy-driven air movement, reducing temperature compared to the flat case.
- Sunlight can reflect between the two panels, allowing **partial utilization of diffuse and reflected radiation**.
- However, at certain sun angles, **one panel may shade the other**, and some radiation passes through the structure.

Impact:

- Moderate temperature reduction
- Improved daily energy yield compared to flat
- Partial airflow and partial solar utilization

→ Performance is **better than flat**, but not optimal.

3. Inverted-V Shape Position PV Panel (Λ)

Description:

In the inverted-V configuration, panels are tilted outward, forming an A-frame or East–West orientation.

Scientific characteristics:

- Both panel backsides are **fully exposed to ambient air**, enabling maximum convective cooling.
- Heated air rises through the center gap, creating a **natural chimney (stack) effect** that continuously removes heat.
- One panel faces east and the other west, allowing **direct sunlight capture during morning and evening hours**.
- This reduces cosine losses and distributes power generation more evenly throughout the day.

Impact:

- Lowest operating temperature
- Best airflow and heat dissipation
- Highest daily energy yield and system efficiency

→ This configuration exhibits the **best overall performance**.